

Efficacy and Safety of Sitagliptin, Metformin and Glimepiride in Type 2 Diabetes: A Phase 3, Double-Blind, Active-Controlled Study

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BACKGROUND

Sitagliptin (DPP-4I) addition to lower dose Glimepiride and Metformin:

- The prolonged activity of intact GLP-1 by DPP-4 inhibition is likely to both, improve glucose-induced insulin secretion and to decrease glucagon levels, resulting in the maintenance of β -cell function¹
- Reduced hypoglycemia events and is weight neutral
- Provides a complementary mechanism for glucose-induced insulin secretion leading to improved diabetes control
- Sitagliptin exhibits known synergism with Metformin through several mechanisms²

Glimepiride high dose can cause:

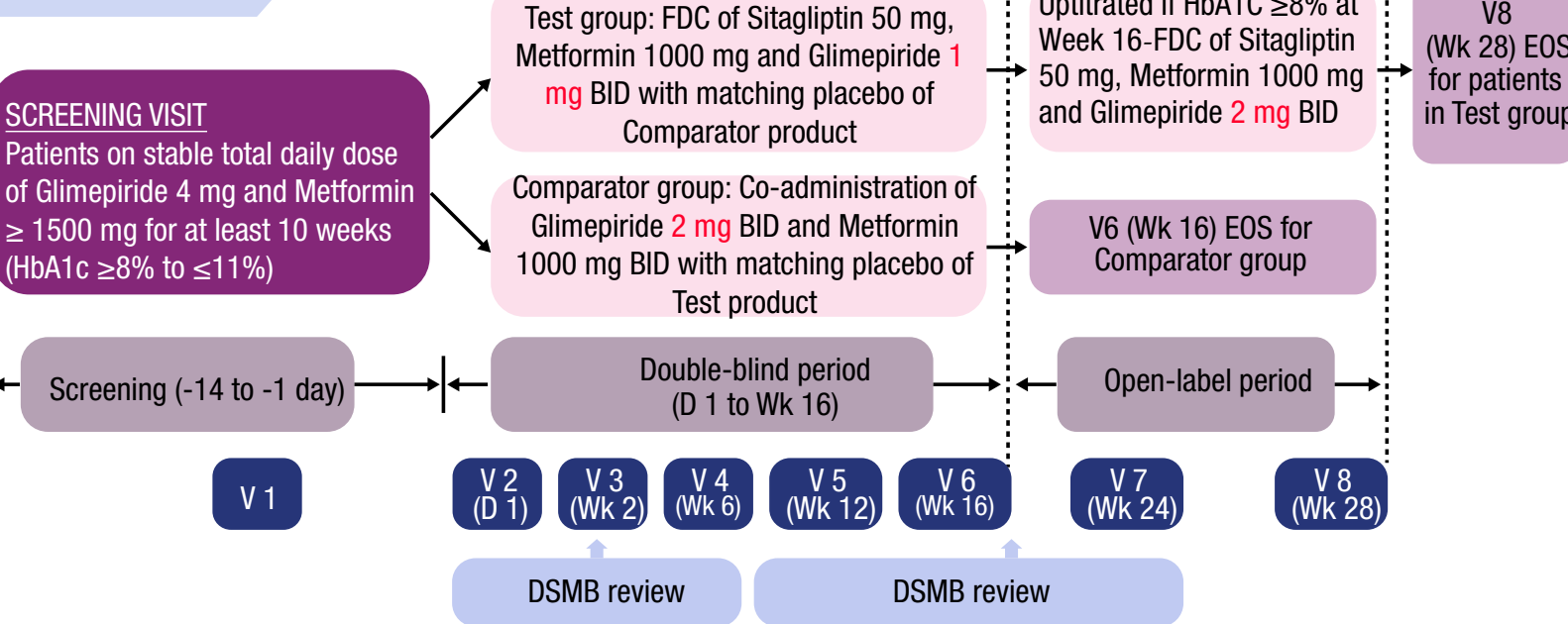
- Progressive β -cell dysfunction
- Higher risk of hypoglycemia events and weight gain
- Efficacy plateau in uncontrolled patients
- Reducing the dose of sulfonylurea and adding DPP-4I may provide better glycemic control without increased risk of hypoglycemia³⁻⁵

A phase 3, randomized, double-blind, double-dummy, active-controlled study (16 weeks) was conducted to evaluate efficacy and safety of Sitagliptin+Glimepiride+Metformin in comparison to co-administration of high dose of Glimepiride and Metformin in patients with T2DM. After 16 weeks, the study was continued as an open-label, single-arm for 12 weeks

METHODS

- This is a phase 3, randomized, double-blind, double-dummy, active-controlled study (16 weeks) followed by an open-label, single-arm study (12 weeks)
- Primary endpoint was change in HbA1c from Baseline to Week 16
- Clinical Trial Registration Number: CTRI/2021/11/038169

Study Design



RESULTS

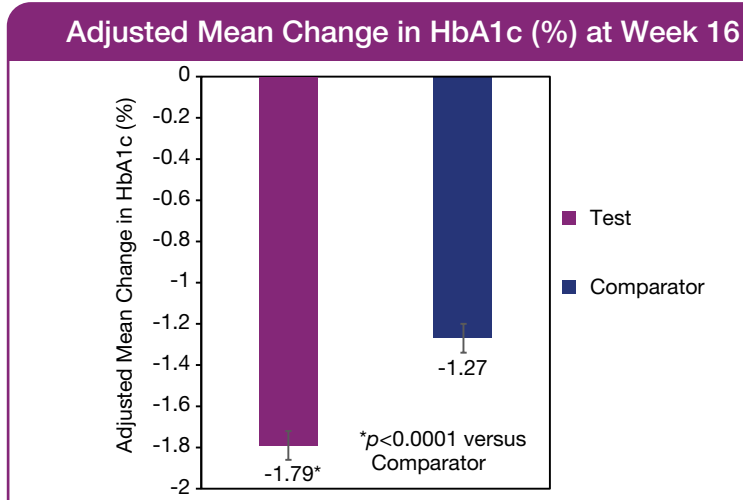
Baseline Demographics

- A total of 392 patients were included in this study. There were 190 patients in Test group and 202 in Comparator group with similar baseline characteristics

	Test group (N=190)	Comparator group (N=202)	Overall (N=392)	p value
Gender n(%)				0.2142
Male	102 (53.7%)	121 (59.9%)	223 (56.9%)	
Female	88 (46.3%)	81 (40.1%)	169 (43.1%)	
Age (Years) Mean $\pm$ SD	47.57 $\pm$ 8.57	47.01 $\pm$ 9.16	47.29 $\pm$ 8.87	0.5339
Body Mass Index (kg/m²) Mean $\pm$ SD	25.49 $\pm$ 3.61	25.95 $\pm$ 3.87	25.73 $\pm$ 3.75	0.2278

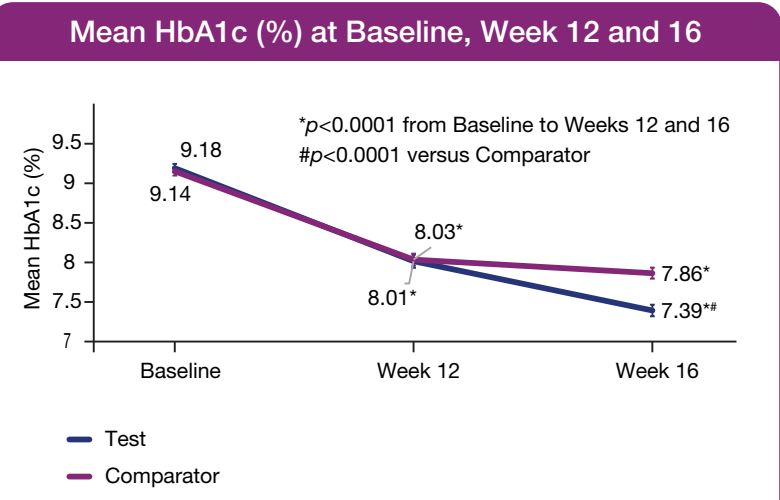
Change in HbA1c (%) at Week 16

- Adjusted mean change (SE) from Baseline in HbA1c (%) at Week 16 was statistically significant in Test group [Test vs Comparator: -1.79 (0.07) vs -1.27 (0.06), $p<0.0001$]



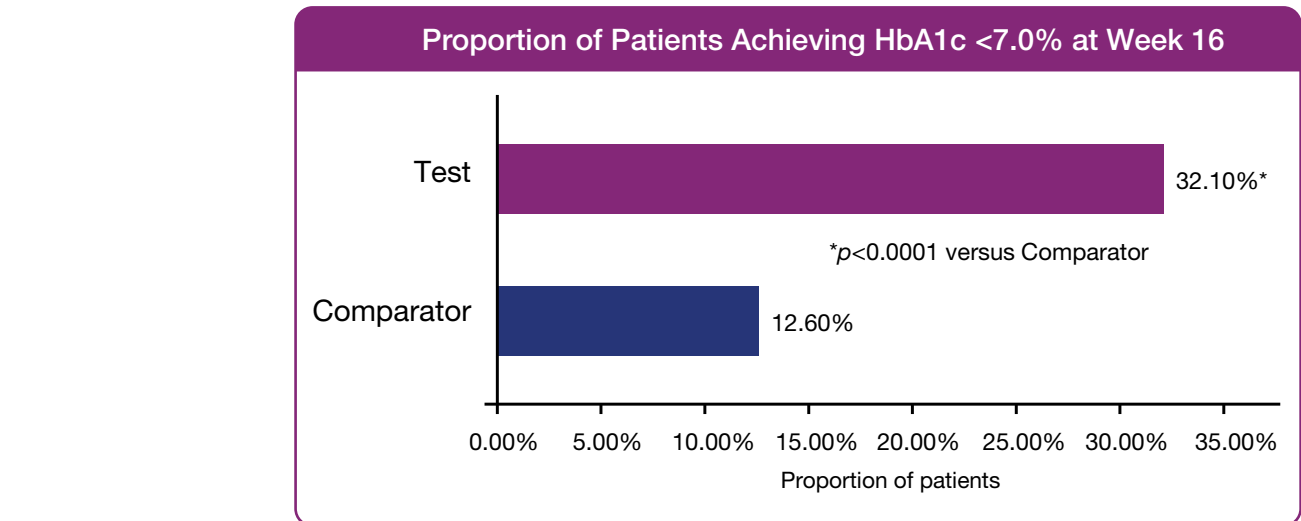
Change in Mean HbA1c (%) from Baseline

- Both treatments demonstrated significant reduction in mean HbA1c (%) from Baseline to Week 12 and Week 16 ($p<0.0001$ each).
- Mean change from Baseline in HbA1c (%) at Week 16 was statistically significant in Test group ($p<0.0001$)



Proportion of Patients Achieving HbA1c <7.0%

- Proportion of patients achieving HbA1c <7.0% was statistically significant at Week 16 (Test vs Comparator: 60 (32.1%) vs 25 (12.6%), $p<0.0001$)



Change in FBG (mg/dL) and PPBG (mg/dL) from Baseline

- Significant reduction in FBG (Test versus Comparator: -32.6 $\pm$ 37.54 versus -35.6 $\pm$ 37.89 and -41.1 $\pm$ 38.97 versus -40.2 $\pm$ 40.43 at Weeks 12 and 16, respectively; $p<0.0001$ each) and PPBG (Test versus Comparator: -41.8 $\pm$ 53.00 versus -44.7 $\pm$ 59.99 and -56.7 $\pm$ 51.80 versus -59.7 $\pm$ 59.96 at Weeks 12 and 16, respectively; $p<0.0001$ each) was noted in both study groups

Change in Glycemic Parameters from Baseline to Week 24 and 28

- Test group demonstrated significant reduction in HbA1c, FBG and PPBG from Baseline to Weeks 24 and 28 ($p<0.0001$ each)

	Baseline (n=182)	Week 24 (n=182)	Week 28 (n=181)
HbA1c (%) (Mean$\pm$SD)	9.16 $\pm$ 0.74	7.06 $\pm$ 1.01	6.79 $\pm$ 1.03
		CFB: -2.10 $\pm$ 0.99	CFB: -2.37 $\pm$ 1.04
		$p<0.0001$	$p<0.0001$
FBG (mg/dL) (Mean$\pm$SD)	176.7 $\pm$ 38.04	128.2 $\pm$ 26.79	123.2 $\pm$ 25.95
		CFB: -48.4 $\pm$ 40.75	CFB: -53.6 $\pm$ 41.74
		$p<0.0001$	$p<0.0001$
PPBG (mg/dL) (Mean$\pm$SD)	251.1 $\pm$ 53.91	178.3 $\pm$ 35.62	172.4 $\pm$ 38.64
		CFB: -72.8 $\pm$ 55.60	CFB: -78.6 $\pm$ 57.25
		$p<0.0001$	$p<0.0001$

Adverse Events: Preferred term – Double Blind Treatment Period

System Organ Class Preferred Term	Test arm (N=190)		Comparator arm (N=202)		Overall (N=392)	
	n (%)	E	n (%)	E	n (%)	E
Total TEAEs	25 (13.2%)	31	30 (14.9%)	36	55 (14.0%)	67
Abdominal pain upper	1 (0.5%)	1	3 (1.5%)	3	4 (1.0%)	4
Constipation	0	0	1 (0.5%)	1	1 (0.3%)	1
Diarrhea	3 (1.6%)	3	5 (2.5%)	6	8 (2.0%)	9
Gastritis	0	0	2 (1.0%)	2	2 (0.5%)	2
Nausea	0	0	6 (3.0%)	6	6 (1.5%)	6
Vomiting	4 (2.1%)	5	4 (2.0%)	4	8 (2.0%)	9
Pyrexia	3 (1.6%)	3	5 (2.5%)	6	8 (2.0%)	9
Nasopharyngitis	1 (0.5%)	1	1 (0.5%)	1	2 (0.5%)	2
Hypoglycemia	3 (1.6%)	5	1 (0.5%)	2	4 (1.0%)	7
Arthralgia	1 (0.5%)	1	0	0	1 (0.3%)	1
Back pain	0	0	1 (0.5%)	1	1 (0.3%)	1
Myalgia	4 (2.1%)	4	0	0	4 (1.0%)	4
Headache	6 (3.2%)	6	3 (1.5%)	3	9 (2.3%)	9
Cough	1 (0.5%)	1	1 (0.5%)	1	2 (0.5%)	2
Rhinorrhea	1 (0.5%)	1	0	0	1 (0.3%)	1

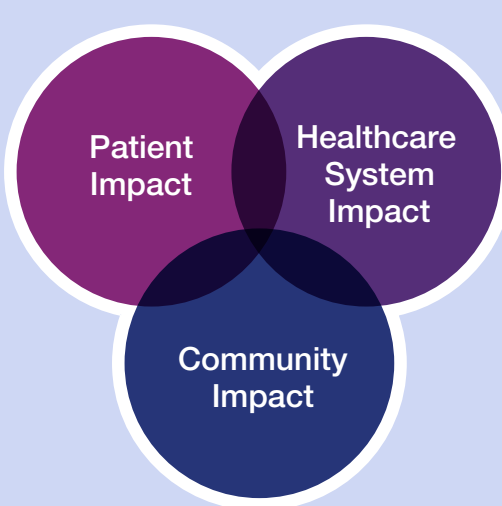
Adverse Events: Preferred term – Open Label Treatment Period

- Total 8 TEAEs were reported during the open label treatment period. These adverse events included pyrexia (n=2), abdominal pain upper, diarrhea, vomiting, hypoglycemia, headache and rhinorrhea (n=1 each)
- All hypoglycemia events were of level 1 and did no require management
- No patient required dose discontinuation in the study

DISCUSSION & CONCLUSION

- Better HbA1c** control at lower dose of Glimepiride
- Less deterioration of pancreatic β cell function** through β cell preservation achieved due to lower Glimepiride dose
- Benefit without increased risk** of hypoglycemia
- Improved compliance

- All study medications were safe and well tolerated



- Decrease in healthcare resource utilization and strain on healthcare system** by slowing disease progression (by β cell exhaustion), more patients achieving target HbA1c and thereby reducing/delaying long-term complications
- Decrease in economic impact, improved productivity of society and better quality of life

Abbreviations: BID: Twice daily; CFB: Change from baseline; D: Day; DPP-4I: Dipeptidyl Peptidase IV Inhibitors; DSMB: Data safety monitoring board; E: Number of Events; EOS: End of study; FBG: Fasting blood glucose; FDC: Fixed dose combination; GLP-1: Glucagon-like peptide-1; HbA1c: Glycated hemoglobin; N: Total number of patients; n: Number of patients; PPBG: Postprandial blood glucose; SD: Standard deviation; T2DM: Type 2 diabetes mellitus; V: Visit; Wk: Week
References: 1. Sato A et al. J Clin Med Res 2019; 11(1): 15 – 20. 2. Janumet – UK Prescribing information, 3. J Clin Med Res. 2014;6(2):127-132; 4. Endocrine Journal 2014, 61 (12), 1163-1170. 5. J Clin Med Res. 2019;11(1):15-20.